

Data is public, available via 'Open Electricity', 'ABS', 'BOM' and 'RBA'. Report for portfolio purposes only. Created by David Prichard

Report	Description
Data Definitions	Has information about each of the reports in this file.
R1 - Electrical Supply Forecast	Shows the 12 month forecasts for Net Electricity Supply (GwH). Contains a forecast for VIC, NSW and QLD.
R2 - Generation Source Proportions	Shows where electricity is sourced from, for example Solar vs Coal. Shows the proportion of source changed over time. Smoothing done using Arithmetic Avg and Exponential Smoothing.
R3 - Generation Source Half Hourly	Shows a typical summer day and the generation source, from the last 28 days

Date Selector - Effects R1 & R2 Reports

1/12/1998



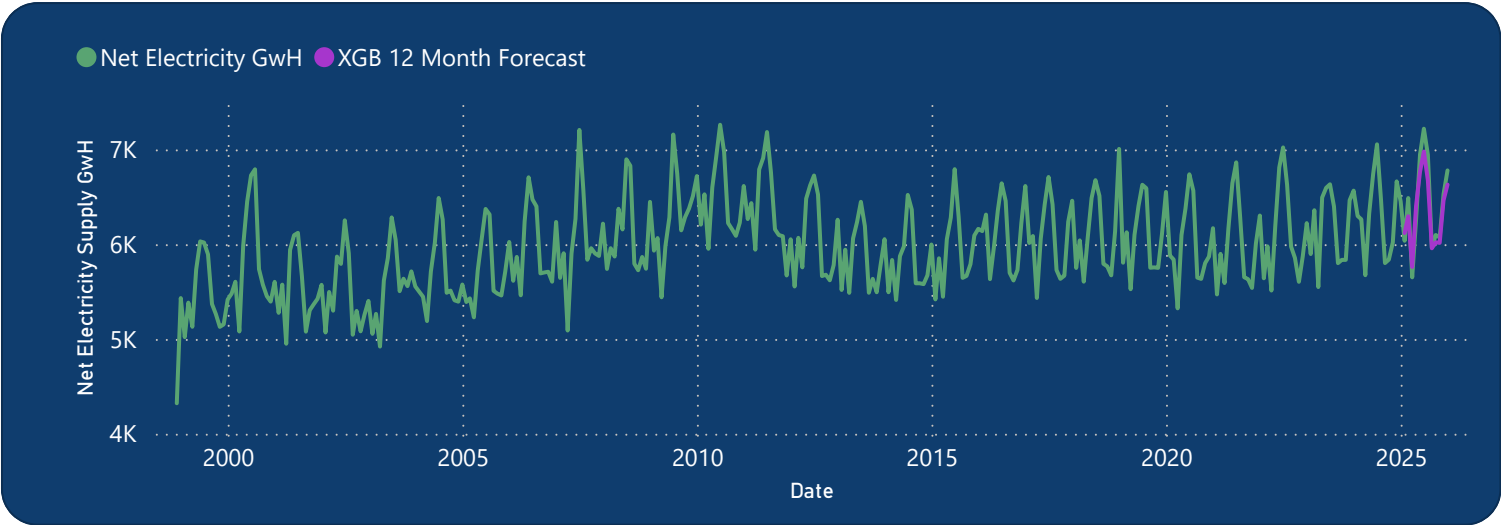
1/01/2026



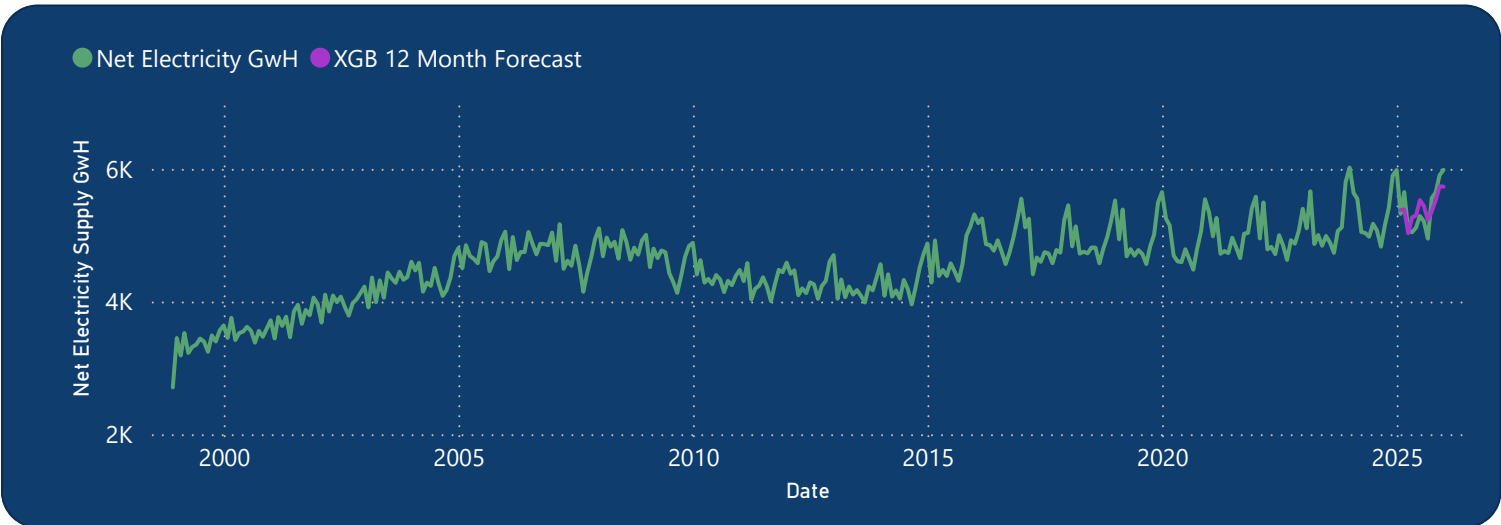
Example Report:

R1 - Electrical Supply Forecast

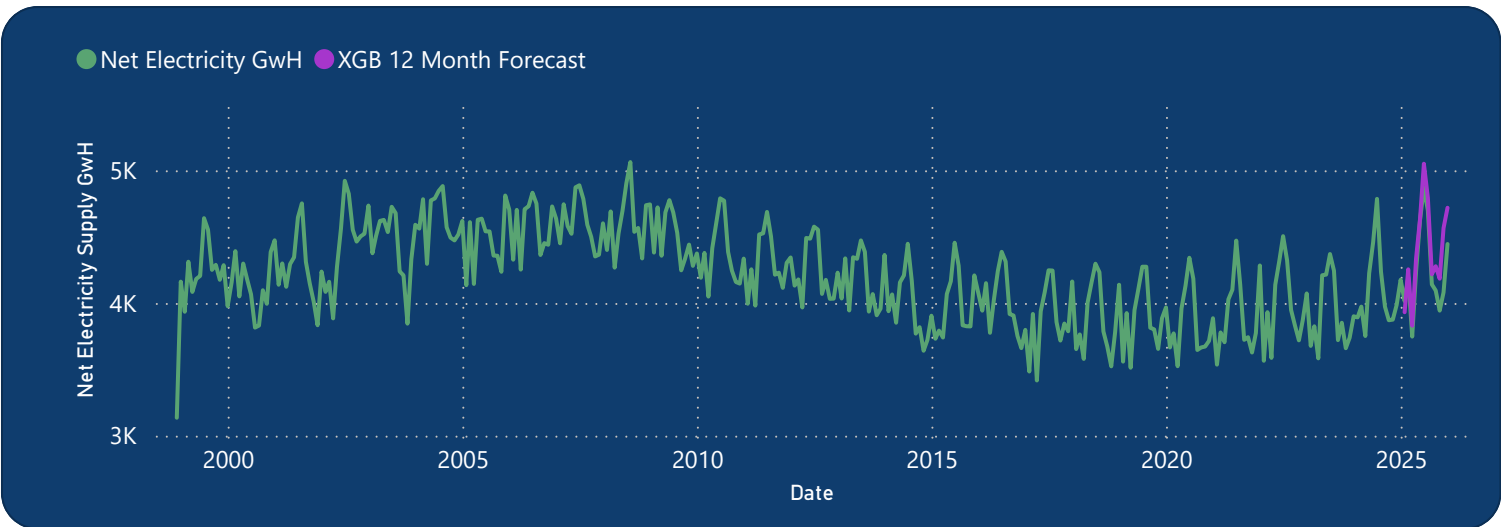
New South Wales



Queensland



Victoria

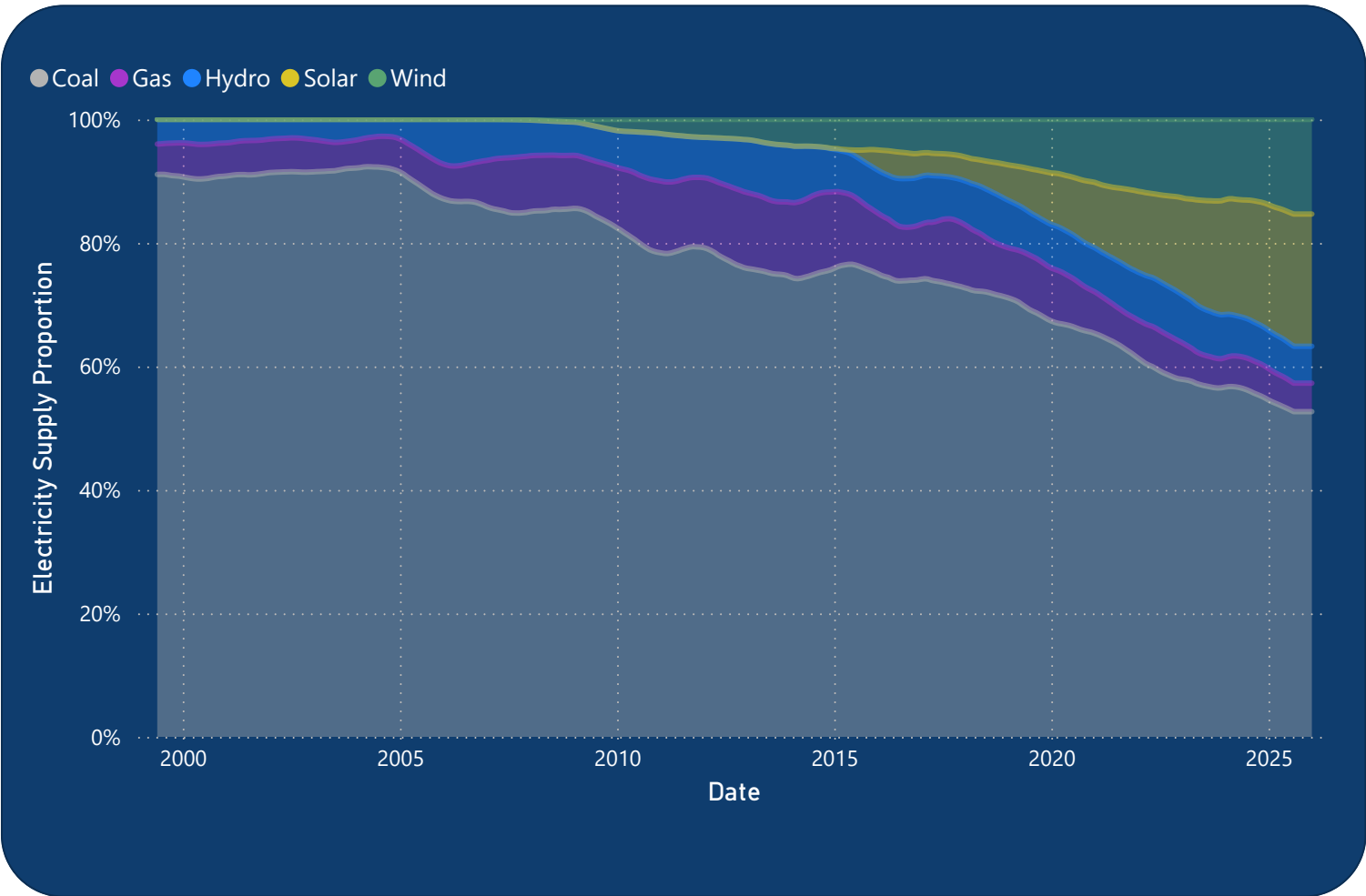
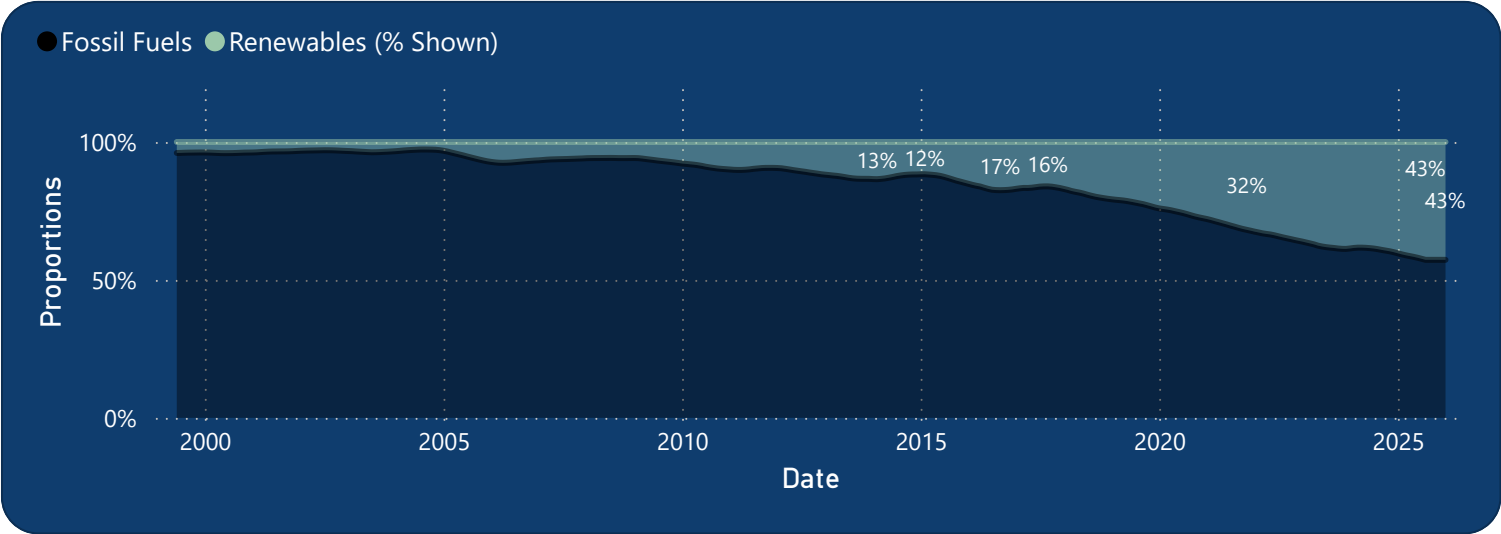


Example Report: R2 - Generation Source Proportions

Recent Proportion - as at January 2026

Coal	Solar	Wind	Hydro	Gas		
52.7%	21.4%	15.3%	6.0%	4.6%	Fossil Fuel	Renewables
					57.3%	42.7%

Proportion Over Time - Smoothed



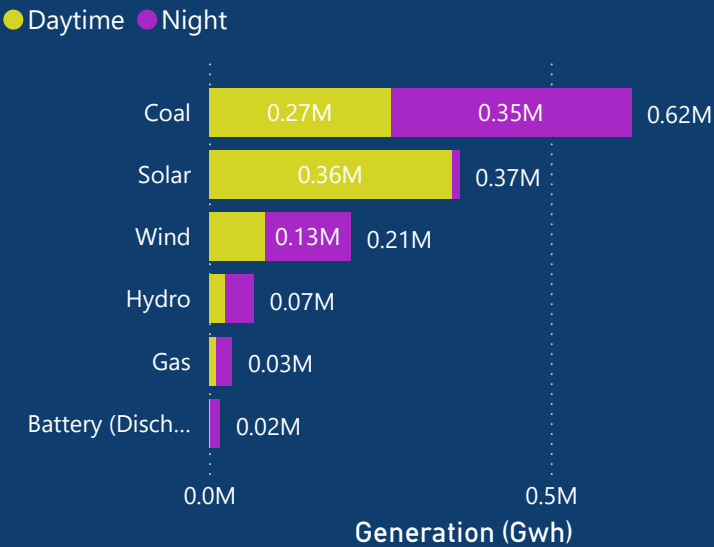
Example Report: R3 - Generation Source Half Hourly

All figures only include the last 28 days as at January 2026

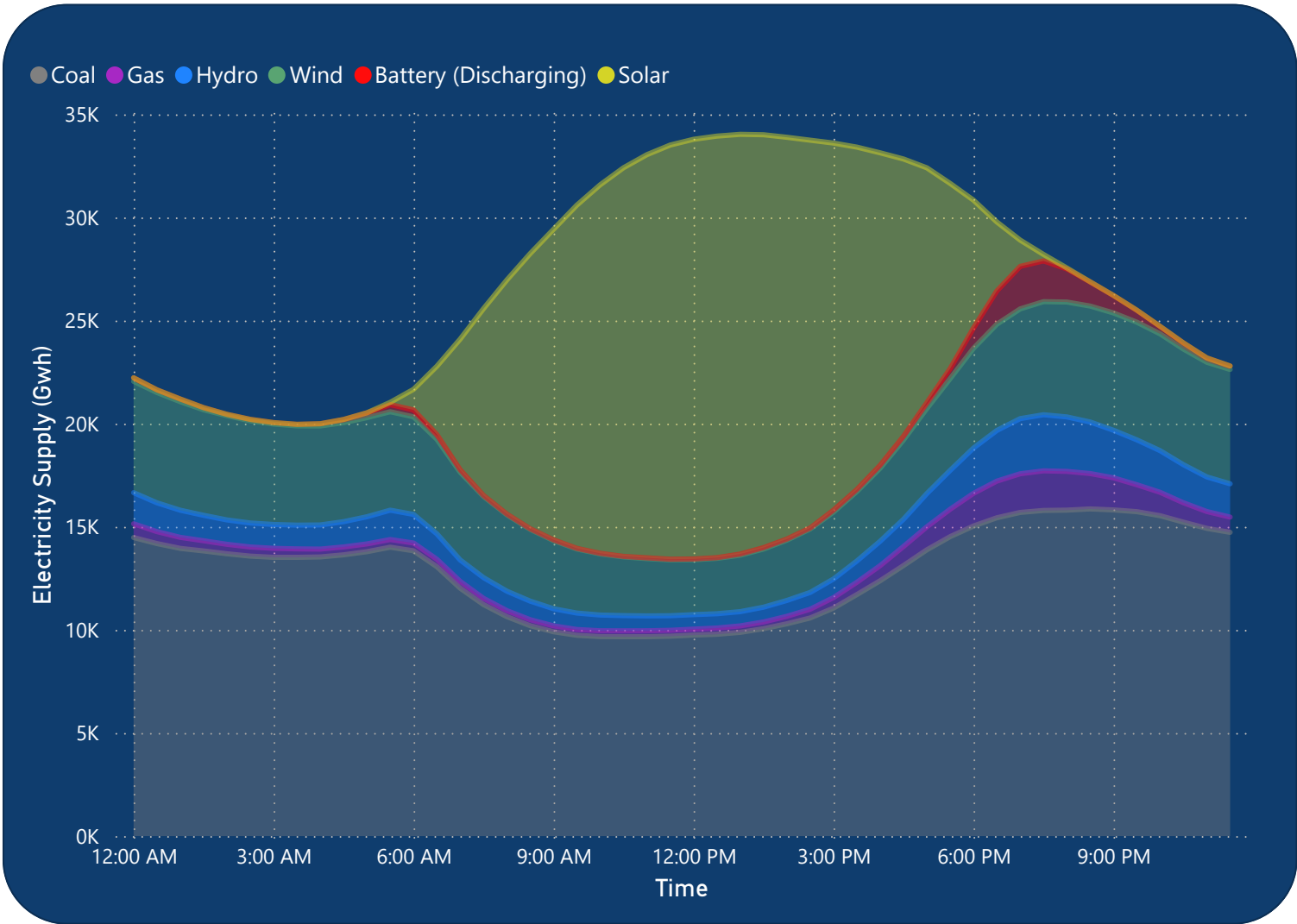
Peak Production Time by Source

Generation Source	Peak Production Time
Solar	12:30 PM
Battery (Discharging)	7:00 PM
Gas	7:30 PM
Hydro	7:30 PM
Coal	8:30 PM
Wind	9:30 PM

Produced by Source and Daytime/Nighttime



Electricity Supply Representative 24h - Averaged over last 28 days



*Summer Period - Naturally High Solar

Example Report: Data Definitions



R1 - Electrical Supply Forecast

Net Electricity GwH

Shows Net Electricity is the actual Gigawatt Hour production for the respective states, this data was provided by <https://openelectricity.org.au/> which sources their data from National Electricity Market (NEM). Net Electricity is calculated by summing the electricity generation over their sources such as Coal or Solar. Note solar does include both commercial and rooftop, rooftops are estimated and does not include wasted solar to grid.

XGB 12 Month Forecast

Shows the estimated 12 month forecast, starting 12 months prior to the last recoded month in the system, currently January 2026. This is so the actual values can be compared to the forecasts. The model uses various data to make the forecast. Given that the 12 month forecast was the goal, there were no weather features or economic features included, these are however in the python notebooks and can be included if for example you wanted a more accurate one month ahead forecast. The model itself is a Extreme Gradient Boosting Tree model, that is high fitting, it uses test-train splits, and is not fed any data within the 12 month forecasting period. Below are the features used and features that can be added if looking at a shorter forecasting horizon, and where you can source weather forecasts for. Features are simply the other variables used to predict future Net Electricity Generation. The forecasts were reasonably predictive, explaining over 80% of the variance in future Net Electricity generation.

Features - 12 Month Forecast, in order of importance:

- State Population - Easy to forward predict
- State Indicator - VIC, NSW, QLD
- t - time in days since first observation
- Trigonometry 2 Period Season - Variable for that is high during peak summer and winter, and low in Autumn and Spring.
- Year^3 - Year Cubed - Used for non-linear t relations
- Trigonometry 1 Period Season - Variable for that is low during peak summer and high during peak winter.
- Year^2 - Year Squared - Used for non-linear t relations

Ad. Features - 1 Month Forecast, in order of importance:

- Monthly Average Max Temperature - Can be estimated a month in advance by weather forecasts.
- Solar Exposure - A proxy can be estimated a month ahead using weather forecasts.
- Trade Weighted AU Dollar - Can be estimated by using the last month.
- While not included an Auto Regressive AR(1) feature would almost certainly improve forecasts for month ahead.

R2 - Generation Source Proportions

Proportion Generation

Shows the proportion of electricity generated by each of the included sources, noting that some smaller insignificant sources were excluded, such as distillate and bioenergy. Solar is combined utility and rooftop. The proportions are a percentage of the total generation that each source contributes in a month. Fossil fuels and renewables are as follows, fossil fuels are coal and gas, renewables are solar, wind and hydro. There should be KPIs for the most recent proportion in the dataset as well as the proportion as it changes over time. The series was smoothed using a 12 month arithmetic (centred window) moving average and an exponential smoothing algorithm over the top.

R3 - Generation Source Half Hourly

Hourly Generation

Generation data is aggregated into half-hours across a 24 hour representative day. The generation data is the last 28 days around the same date as the most recent generation data is taken, so January 2026. Noting that if this is during summer then the generation for solar will look very different to winter.

Daytime/Night Generation

The daytime vs night chart is net production over the period of 28 days broken down into generation daytime between 6am-6pm else it is considered night.

Peak Production by Source

Shows the peak production time for each of the generation sources. Should show when each electricity source, such as solar produces the most electricity. Noting that many sources can be turned off and on, while some sources are more constant like coal. Coal produces electricity all through the night largely to heat hot water, since coal cannot be turned off quickly.

Additional Notes

All data sources are linked in the below notebooks:

- [electricity_demand_analysis.ipynb](#)
- [supply_electricity_forecasting_features.ipynb](#)
- [australia_timeseries_plots_analysis.ipynb](#)

All data is linked to parquet files in the Portfolio Output Folder